
Divyansh Nilesh Jain

Academic Portfolio — 2026

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Hi, I'm *Divyansh* Jain

First-year B.Tech student at MITAOE, pursuing CSE with specialisation in AI & ML. Passionate about data analytics, visualisation, and emerging technologies — turning ideas into meaningful solutions, one project at a time.

VIEW MY WORK

GET IN TOUCH



Divyansh Jain

MITAOE · B.Tech CSE (AIML)

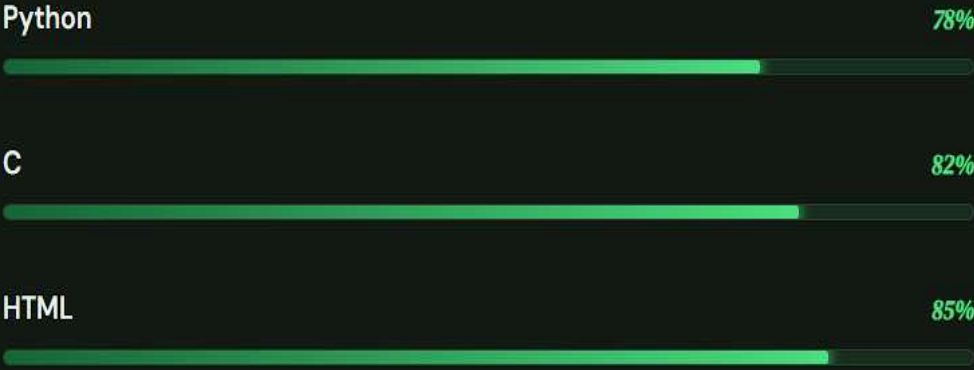


1st
YEAR B.TECH

2+
PROJECTS BUILT

3+
TECH SKILLS

01 Skills



02 What I Do



FRONTEND DEV

Building responsive, performant interfaces with modern frameworks and meticulous attention to user experience.



BACKEND DEV

Scalable APIs and services built with a focus on reliability, security, and clean architecture.



DATA VISUALIZATION

Using Power BI to clean datasets and craft insightful dashboards that tell a clear, compelling story.



03 Projects

Smart Helmet Lite

HARDWARE · ARDUINO · IOT

Smart Helmet Lite

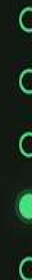
Designed a smart helmet to protect workers and reduce occupational stress. Built using Arduino and microcontrollers, integrating sensors to monitor safety conditions in real time.

Smart Pavement Light

HARDWARE · C++ · ARDUINO · ENERGY

Smart Pavement Light

Made "Turning footsteps into electricity" a reality. Programmed in C++ on Arduino, converting mechanical energy from footsteps into electrical energy to power pavement lights.



04 Let's Connect

"Open to collaborations, interesting projects, and new opportunities. Whether you want to build something cool or just have a chat — reach out!"

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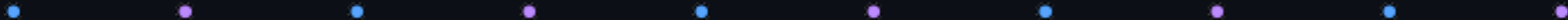
Creative Technologies

Lab 1–7: HTML, CSS & Digital Portfolio

Name: Divyansh Nilesh Jain

PRN: 202501110088

Class: AI1



Objective

To study the fundamentals of Web Technology and implement basic HTML concepts including structure, tags, lists, images, and hyperlinks. The experiment focuses on understanding the building blocks of web pages and how browsers render HTML content.

Theory

- **WWW** is a system of interlinked web pages accessed via the Internet.
- **Internet** is the network; **WWW** is a service running on top of it.
- **HTTP** (HyperText Transfer Protocol) enables communication between browsers and servers.
- **URL** (Uniform Resource Locator) uniquely identifies each webpage on the internet.
- **Browsers** parse and render HTML to display webpages visually.

HTML Introduction

HTML (HyperText Markup Language) is the standard language used to create and structure content on the web. It uses **tags** enclosed in angle brackets to define elements like headings, paragraphs, links, and images.

Basic Structure: `<html>` contains the entire document; `<head>` holds metadata and title; `<body>` contains all visible content rendered in the browser.

Program 1: Basic HTML Page

index.html

```
<!DOCTYPE html>
<html>
<head>
<title>My First Page</title>
</head>
<body>

<h1>Welcome to My Website</h1>
<p>This is my first HTML page.</p>

</body>
</html>
```

file:///C:/Users/output.html

Welcome to My Website

This is my first HTML page.

Program 2: Ordered & Unordered Lists

lists.html

```
<!DOCTYPE html>
<html>
<head>
<title>List Example</title>
</head>
<body>

<h2>Ordered List</h2>
<ol>
  <li>Morning Walk</li>
  <li>Study</li>
  <li>Coding</li>
</ol>

<h2>Unordered List</h2>
<ul>
  <li>Music</li>
  <li>Sports</li>
  <li>Gaming</li>
</ul>

</body>
</html>
```

file:///C:/Users/lists.html

Ordered List

1. Morning Walk
2. Study
3. Coding

Unordered List

- Music
- Sports
- Gaming

link_image.html

```
<html>
<head>
<title>Link and Image</title>
</head>
<body>

<h2>My College Link</h2>
<a href="https://mitaoe.ac.in"
  target="_blank">
  Visit MITAOE
</a>

<h2>My Image</h2>


</body>
</html>
```

file:///C:/Users/link_image.html

My College Link

[Visit MITAOE](https://mitaoe.ac.in)

My Image

[Image Loaded
from Google Drive]

What We Achieved

✓ HTML Structure

Gained hands-on experience with the fundamental structure of an HTML document — DOCTYPE, html, head, and body tags — and understood how browsers parse and render these elements.

🔗 Hyperlinks & Navigation

Successfully implemented anchor tags with href attributes and target='_blank' to open external links in new tabs, understanding the role of hyperlinks in web navigation.

🖼️ Images in Web Pages

Embedded images using the img tag with src, alt, and width attributes, and explored using external URLs (Google Drive) as image sources in HTML pages.

📁 Lists & Organization

Created both ordered (numbered) and unordered (bulleted) lists using ol, ul, and li tags, demonstrating how HTML structures content for readability and accessibility.

Objective

To study and implement advanced HTML concepts including:

- Hyperlink and image tags with attributes
- Text formatting tags (b, i, u, strong, em)
- Block-level div and inline span elements
- Iframe for embedding external content
- Multimedia elements: audio and video
- HTML forms for user input collection

Theory

HTML provides a rich set of elements beyond basic structure. Formatting tags like b, i, u, strong, and em control text appearance. The div tag creates block-level containers for layout, while span allows inline styling within text.

The iframe element embeds external web pages or resources directly within a page. HTML5 introduced native audio and video tags enabling multimedia without plugins. Forms enable interactive data collection using input elements of various types.

Prog 1

Formatting + Div + Span

Prog 2

Image + Link

Prog 3

Iframe

Prog 4

Video + Audio

Prog 5

HTML Form

010

| Creative

Program 1: Formatting + Div + Span

formatting.html

```
<!DOCTYPE html>
<html>
<head>
<title>Formatting Example</title>
</head>
<body>

<h1>Text Formatting</h1>
<b>Bold Text</b><br>
<i>Italic Text</i><br>
<u>Underline Text</u><br>
<strong>Strong Text</strong><br>
<em>Emphasized Text</em>

<div style="color:blue;
font-size:20px;">
This is a DIV section
</div>

<p>This is <span
style="color:red;">
SPAN text</span></p>

</body>
</html>
```

file:///C:/Users/formatting.html

Text Formatting

Bold Text

Italic Text

Underline Text

Strong Text

Emphasized Text

This is a DIV section

This is ~~SPAN~~ text

Program 2: Image + Link

image_link.html

```
<!DOCTYPE html>
<html>
<body>

<h2>Image and Link</h2>

<a href="https://mitaoe.ac.in"
  target="_blank">
  Visit MITAOE
</a>

<br><br>

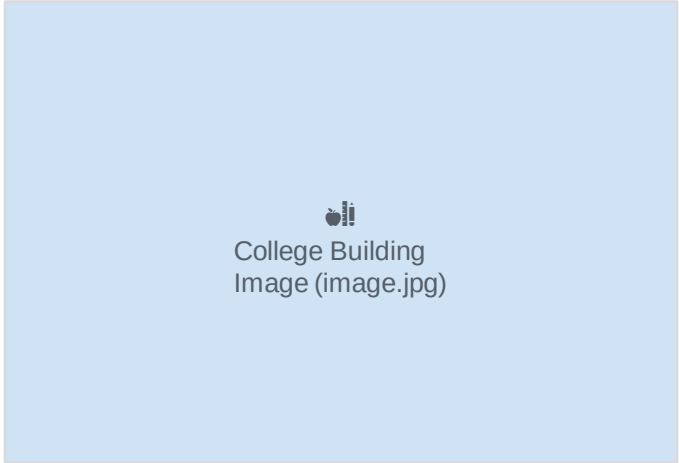


</body>
</html>
```

file:///C:/Users/image_link.html

Image and Link

[Visit MITAOE](https://mitaoe.ac.in)



College Building
Image (image.jpg)

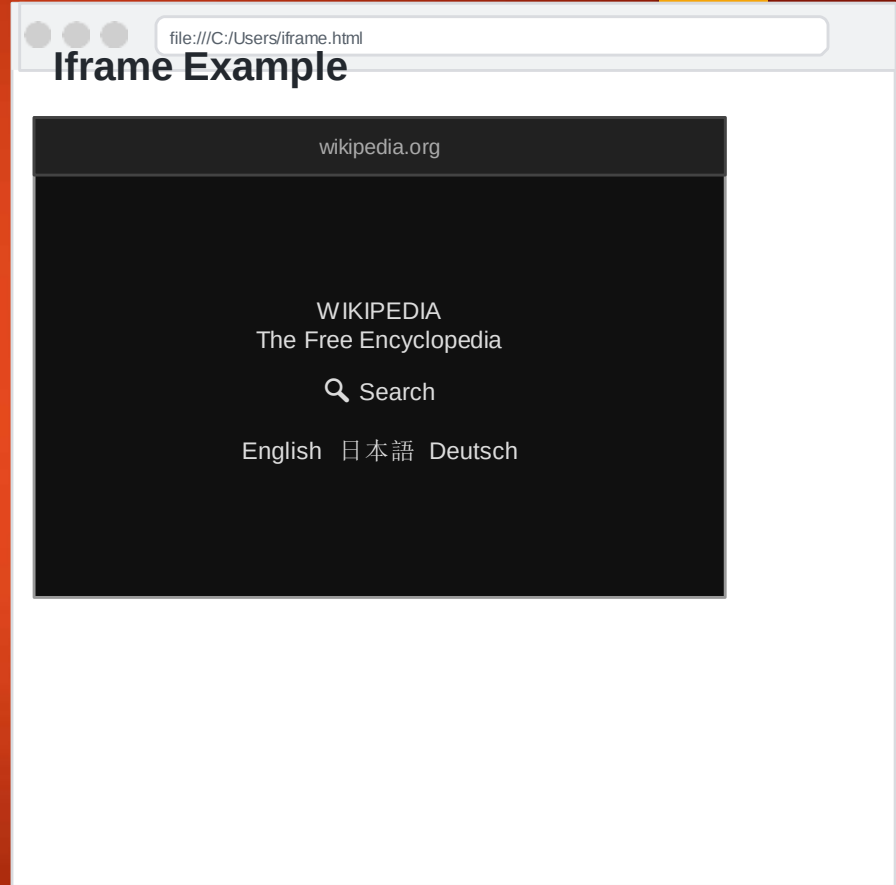
Program 3: Iframe

```
iframe.html
<!DOCTYPE html>
<html>
<body>

<h2>Iframe Example</h2>

<iframe
  src="https://www.wikipedia.org"
  width="300"
  height="200">
</iframe>

</body>
</html>
```



Program 4: Video + Audio

media.html

```
<!DOCTYPE html>
<html>
<body>

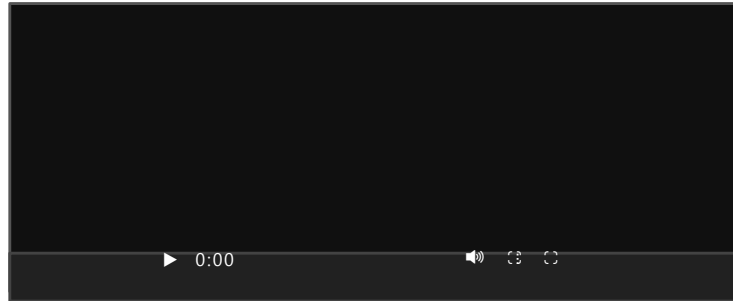
<h2>Video Example</h2>
<video width="300" controls>
  <source src="video.mp4"
    type="video/mp4">
</video>

<h2>Audio Example</h2>
<audio controls>
  <source src="audio.mp3"
    type="audio/mpeg">
</audio>

</body>
</html>
```

file:///C:/Users/media.html

Video Example



Audio Example



●●● Program 5: HTML Form

```
form.html

<!DOCTYPE html>
<html>
<body>

<h2>Registration Form</h2>

<form>
  Name:
  <input type="text"> <br> <br>

  Email:
  <input type="email"> <br> <br>

  Password:
  <input type="password"> <br> <br>

  DOB:
  <input type="date"> <br> <br>

  <input type="submit"
    value="Submit">
</form>

</body>
</html>
```

file:///C:/Users/form.html

Registration Form

Name:

Email:

Password:

DOB:

Advanced HTML — Key Takeaways

Text Formatting Mastered

Implemented all major formatting tags (`b`, `i`, `u`, `strong`, `em`) and learned the semantic difference between `strong/em` vs `b/i`. Used `div` for block-level and `span` for inline styling with CSS.

Images & External Links

Embedded images from local paths and explored using relative vs absolute URLs. Linked to external sites using anchor tags with target attributes for controlled navigation.

Multimedia Integration

Used HTML5 native video and audio tags with the controls attribute, and source elements for multiple format fallbacks — enabling media playback without any external plugins.

Forms for Interactivity

Built a registration form with text, email, password, and date input types, understanding how form elements capture user data for submission to web servers.

Cascading Style Sheets

Objective

To study and implement Cascading Style Sheets (CSS) including inline, internal, and external CSS for designing visually appealing web pages. The lab covers CSS selectors, the box model, layout design, and the proper use of class and ID selectors.

Theory

CSS (Cascading Style Sheets) controls the visual presentation of HTML elements — including colors, fonts, spacing, and layout. There are three ways to apply CSS:

- Inline CSS: Applied directly to an element via the style attribute.
- Internal CSS: Written inside a <style> tag in the HTML <head>.
- External CSS: Stored in a separate .css file and linked via <link> tag.

CSS cascades, meaning styles from multiple sources are combined with specificity rules determining priority.

Inline CSS

```
style="color:red"
```

Internal CSS

```
<style> in <head>
```

External CSS

```
<link href="style.css">
```

Program 1: Inline CSS

inline.html

```
<!DOCTYPE html>
<html>
<head>
<title>Inline CSS</title>
</head>
<body>

<p style="color:green;
font-size:30px;
text-align:center;">
This is Inline CSS Example
</p>

</body>
</html>
```

file:///C:/Users/inline.html

This is Inline CSS Example

Program 2: Internal CSS

internal.html

```
<!DOCTYPE html>
<html>
<head>
<title>Internal CSS</title>
<style>
body {
  background-color: lightyellow;
}
h1 {
  color: blue;
  text-align: center;
}
p {
  font-size: 20px;
}
</style>
</head>
<body>
<h1>Internal CSS</h1>
<p>This is paragraph using
internal CSS.</p>
</body>
</html>
```

Internal CSS

This is paragraph using internal CSS.

Program 3: External CSS

external.html

```
/* style.css */
body {
  background-color: lightblue;
}
h1 {
  color: green;
  text-align: center;
}
p {
  font-size: 18px;
}
```

```
/* index.html */
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet"
      href="style.css">
</head>
<body>
<h1>External CSS Example</h1>
<p>This is paragraph
    using CSS.</p>
</body>
</html>
```

External CSS Example

This is paragraph using CSS.

Program 4: Div, Class and ID

div_class.html

```
<!DOCTYPE html>
<html>
<head>
<style>
.main {
  text-align: center;
}
.highlight {
  color: red;
  font-size: 25px;
}
#special {
  color: blue;
  font-weight: bold;
}
</style>
</head>
<body>
<div class="main">
  <p class="highlight">
    This is class example
  </p>
  <p id="special">
    This is ID example
  </p>
</div>
</body>
</html>
```

file:///C:/Users/div_class.html

This is class example

This is ID example

Program 5: Page Layout Design

layout.html

```
<!DOCTYPE html>
<html>
<head>
<style>
header {
  background-color: black;
  color: white;
  text-align: center;
  padding: 10px;
}
section {
  background-color: lightgray;
  padding: 20px;
}
footer {
  background-color: black;
  color: white;
  text-align: center;
}
</style>
</head>
<body>
<header>
  <h1>My Website</h1>
</header>
<section>
  <p>This is content area.</p>
</section>
<footer>
  <p>Footer Section</p>
</footer>
</body>
</html>
```

file:///C:/Users/layout.html

My Website

This is content area.

Footer Section

Soham Rajaram Chadge | PRN: 2025010
| Creditive Technologies

CSS — Styling the Web

Three Types of CSS

Mastered inline, internal, and external CSS. Understood that external CSS promotes code reusability across multiple pages while inline CSS offers quick per-element overrides for specific styling needs.

Selectors: Class & ID

Applied class selectors (.) for reusable styles on multiple elements and ID selectors (#) for unique styling. Understood the specificity hierarchy: ID > Class > Element.

Layout with CSS

Designed a structured web page with header, section, and footer using CSS background-color, padding, and text-align properties — demonstrating professional page layout principles.

Visual Web Design

Transformed plain HTML into visually appealing pages using CSS. Learned how separating structure (HTML) from presentation (CSS) makes web development more maintainable and scalable.

Objective

To design and develop a complete digital portfolio website using HTML and CSS, and to present it effectively. The portfolio demonstrates the student's skills, projects, and achievements in a professional online format.

Theory

A digital portfolio is an online platform that showcases personal information, skills, projects, and achievements. It serves as an interactive online resume, helping professionals and students present their work to potential employers or academic institutions.

A website is built using three core technologies: HTML for content structure, CSS for visual presentation and layout, and JavaScript for interactive functionality. Together these enable creation of dynamic, responsive, and professional web experiences.

Portfolio Sections Built

 About Me

 Skills

 Projects

 @Contact

 Styled with CSS

 Responsive Layout

Portfolio: Code & Output

portfolio.html

```
<!DOCTYPE html>
<html>
<head>
<title>Portfolio</title>
<style>
body {
  background: #050b16;
  color: white;
  font-family: Arial;
  text-align: center;
}
h1 { font-size: 40px; }
.card {
  background: #1a2232;
  padding: 15px;
  margin: 10px;
  border-radius: 10px;
}
</style>
</head>
<body>
<h1>Divyansh Jain</h1>
<p>Web Development Learner</p>
<h2>Skills</h2>
<div class="card">HTML</div>
<div class="card">CSS</div>
<div class="card">Java</div>
<h2>Projects</h2>
<div class="card">EduConnect</div>
<div class="card">Data Visualization</div>
<h2>Contact</h2>
<p>Email: divyanshjain0404@gmail.com</p>
</body>
</html>
```



Portfolio: Link & Image – Live Demo

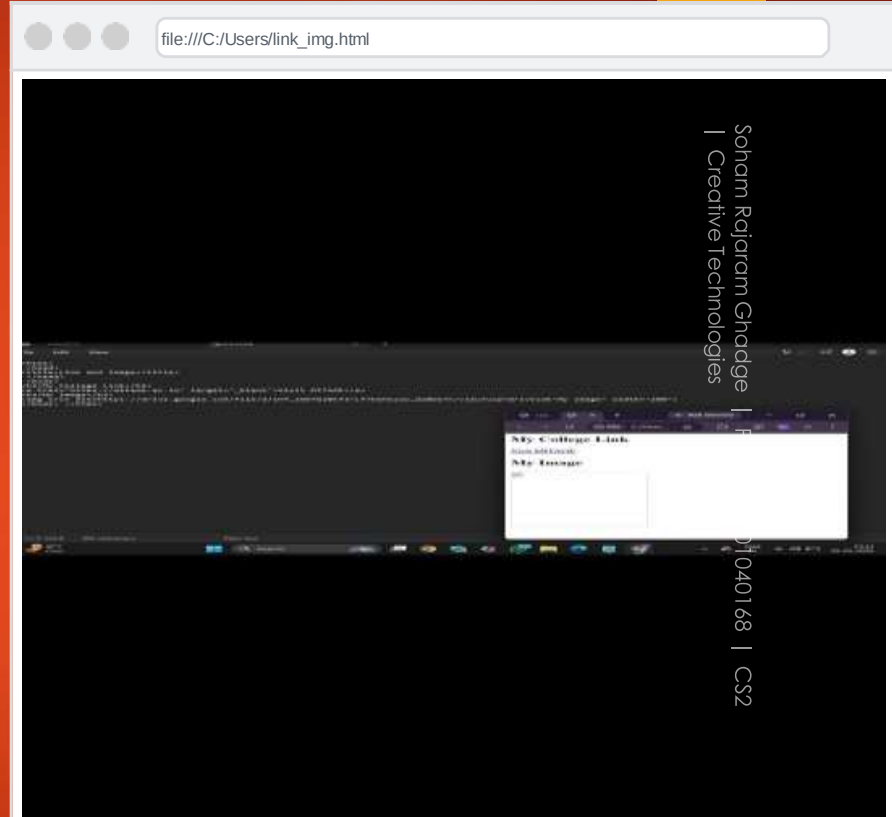
link_img.html

```
<html>
<head>
<title>Link and Image</title>
</head>
<body>

<h2>My College Link</h2>
<a href="https://mitaoe.ac.in"
  target="_blank">
  Visit MITAOE
</a>

<h2>My Image</h2>


</body>
</html>
```



Digital Portfolio — Reflections

End-to-End Web Project

Successfully built a complete, multi-section portfolio website from scratch using only HTML and CSS — demonstrating the ability to independently create and style a functional web project.

Design with CSS Cards

Implemented a dark-themed design using background colors, border-radius card components, and CSS-driven layout — creating a professional, modern aesthetic that mirrors real-world portfolios.

Content Architecture

Organized content into logical sections — About, Skills, Projects, and Contact — understanding how information architecture affects user experience and how portfolios communicate value to viewers.

Presentation & Deployment

Presented the live portfolio demonstrating real browser rendering. The exercise built confidence in creating public-facing web pages and showcasing technical work in an accessible online format.

Objective

To understand the basics of web development and learn the concept of a digital portfolio. This includes comprehending the role of HTML, CSS, and JavaScript in building websites, and recognizing different types of web developers and their responsibilities.

Digital Portfolio

An online platform that showcases personal information, skills, projects, and achievements. It acts as an interactive digital resume, giving potential employers or institutions direct access to a candidate's body of work and capabilities.

Components of a Website

HTML → Structure and content of the page

CSS → Visual styling, layout, and design

JavaScript → Interactivity and dynamic behavior

Types of Web Developers

Frontend - user interface using HTML, CSS, JavaScript

Backend - server logic, databases and APIs

Full stack - on both frontend and backend

Web Development — Core Understanding

Web Architecture Clarity

Gained a clear conceptual understanding of how the web functions — from HTTP requests and URL resolution to how browsers parse HTML, apply CSS, and execute JavaScript to render web pages.

Portfolio as Professional Tool

Understood that a digital portfolio is not just a project — it is a professional asset that communicates skills, creativity, and technical ability to employers, making it essential for career development in tech.

Developer Roles Demystified

Clarified the distinct roles of frontend, backend, and full-stack developers, providing a roadmap for career specialization and helping align personal strengths with appropriate development tracks.

Technology Stack Foundation

Solidified understanding that modern web development is built on three pillars — HTML, CSS, and JavaScript — each serving a specific purpose that together create the complete web experience.

Creative Technologies

Lab Assignment Complete

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Class: AL1

Prompt Engineering Mini Project Report

Project 9: Interview Simulator & Feedback Coach

Tool Used: Ollama (gemma4:31b-cloud)

Date: May 2026

1. Problem Statement

Many students experience anxiety and lack of preparedness before club interviews or internship selection rounds. Without proper practice, they struggle to articulate their technical skills, past projects, and thought processes clearly under pressure. A real-time AI-powered interview simulator can bridge this gap by providing personalized, adaptive practice sessions with constructive feedback — anytime, without needing a human interviewer.

2. Objective

The objective of this mini project is to apply **Role-Playing** and **Iterative Feedback** prompt engineering techniques to create an interactive interview simulator using a local LLM (Gemma 4 via Ollama). The AI is instructed to act as a strict technical interviewer, ask one question at a time, give structured feedback on each answer, and produce a final scorecard.

3. Tools Used

Tool	Purpose
Ollama	Local LLM runtime environment
gemma4:31b-cloud	Language model used for the interview simulation
Windows 11	Operating system / host platform

4. Initial Prompt (Before Refinement)

Prompt Used:

"Interview me for a club."

Output (Screenshot):

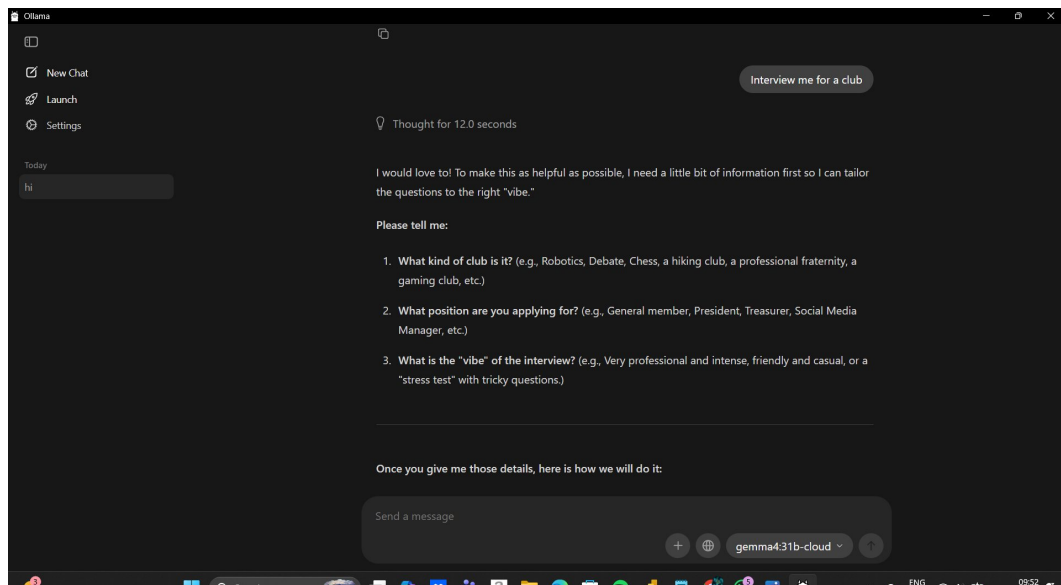


Figure 1 — Initial prompt: AI asks multiple clarifying questions instead of starting the interview

Observation:

The initial prompt was too vague. Instead of starting an interview, the AI asked the user three meta-questions: (1) What kind of club? (2) What position? (3) What tone? This demonstrates that **under-specified prompts lead to clarification loops** rather than actionable outputs. The AI did not take on any interviewer role.

5. Refined Prompt (After Engineering)

Prompt Used:

"Act as a strict but encouraging technical interviewer for the college Robotics Club. Rules you must follow: Ask me ONE question at a time only. After I give my answer, provide constructive feedback in 2-3 lines. Then ask the next question. After 5 questions, give me a final scorecard rating me out of 10 on: Confidence, Clarity, Technical Knowledge, and Communication. Start now by introducing yourself briefly and asking Question 1."

Techniques Applied:

- **Role-Playing:** AI is given a specific persona — 'strict but encouraging technical interviewer'.
- **Conditional Logic:** 'After I give my answer → give feedback → ask next question'.
- **Constraint Setting:** One question at a time, 5 questions total.
- **Structured Output:** Final scorecard with 4 defined criteria.
- **Iterative Feedback Loop:** Each answer triggers evaluation before progression.

Output (Screenshot):

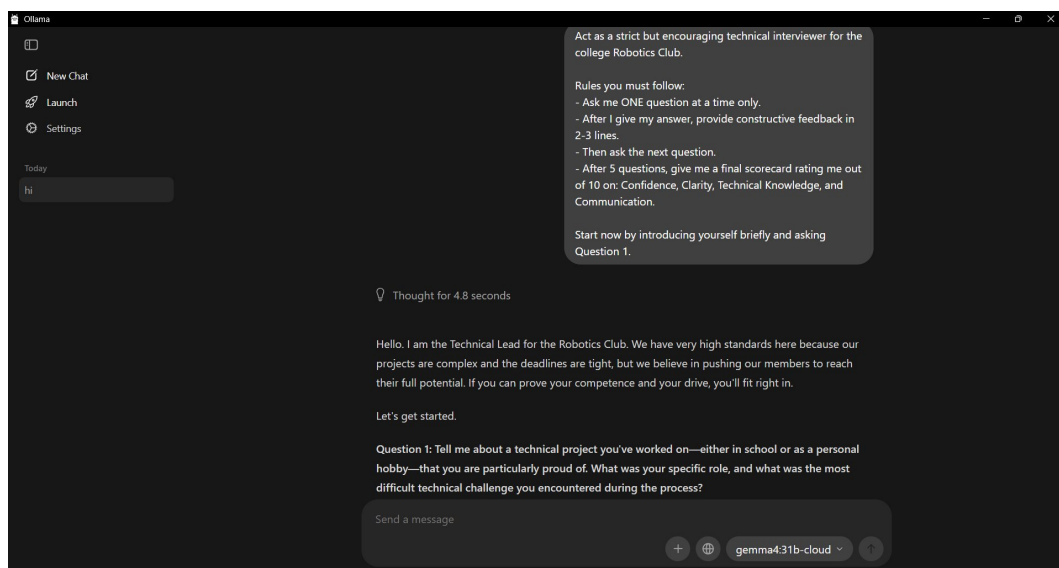


Figure 2 — Refined prompt: AI immediately adopts the interviewer persona and asks Question 1

6. Interview Session — Q&A; Transcript

Question 1 & Feedback

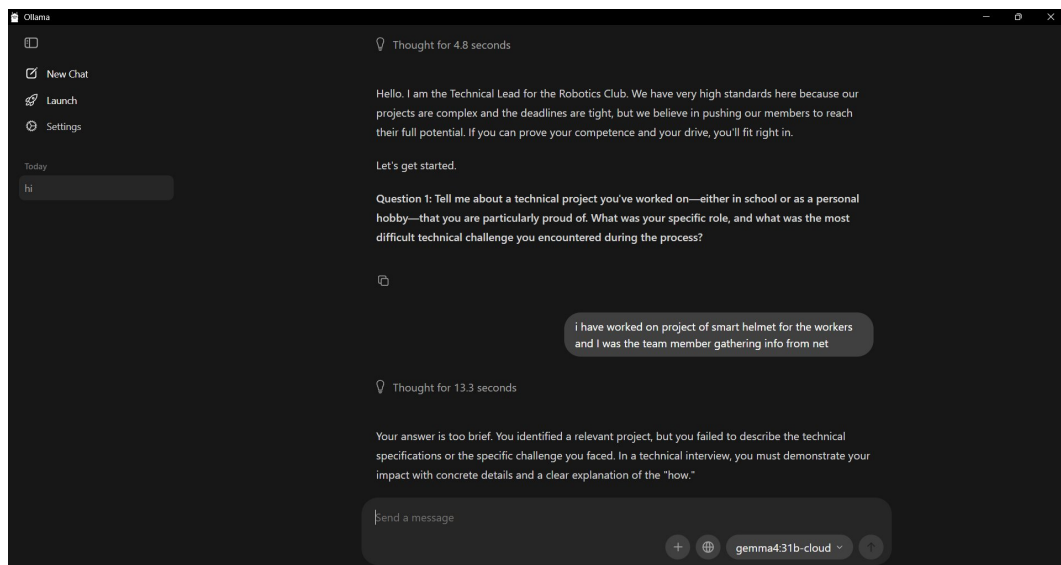


Figure 3 — Q1: AI asks about a technical project; user answers briefly about a smart helmet project

Feedback given by AI: The answer was flagged as too brief. The AI pointed out that the user failed to describe technical specifications or the specific challenge faced, and asked for concrete details about the 'how'.

Question 2 & Feedback

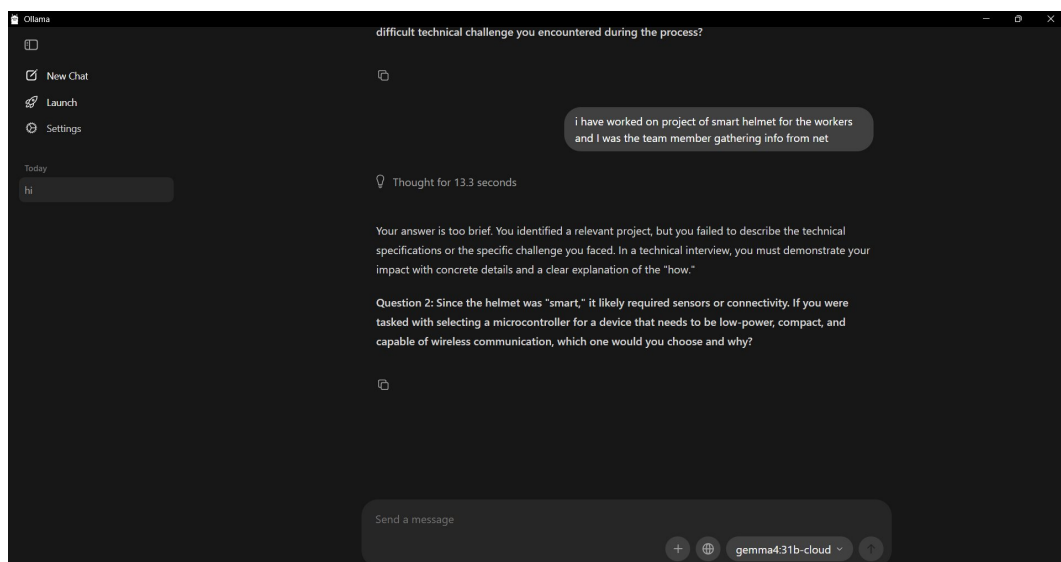


Figure 4 — Q2: AI asks about microcontroller selection; user gives an off-topic answer

Feedback given by AI: The user completely bypassed the microcontroller question and told an unrelated story. The AI correctly identified this as a critical communication error — failing to listen to the specific question asked.

Question 3 & Feedback

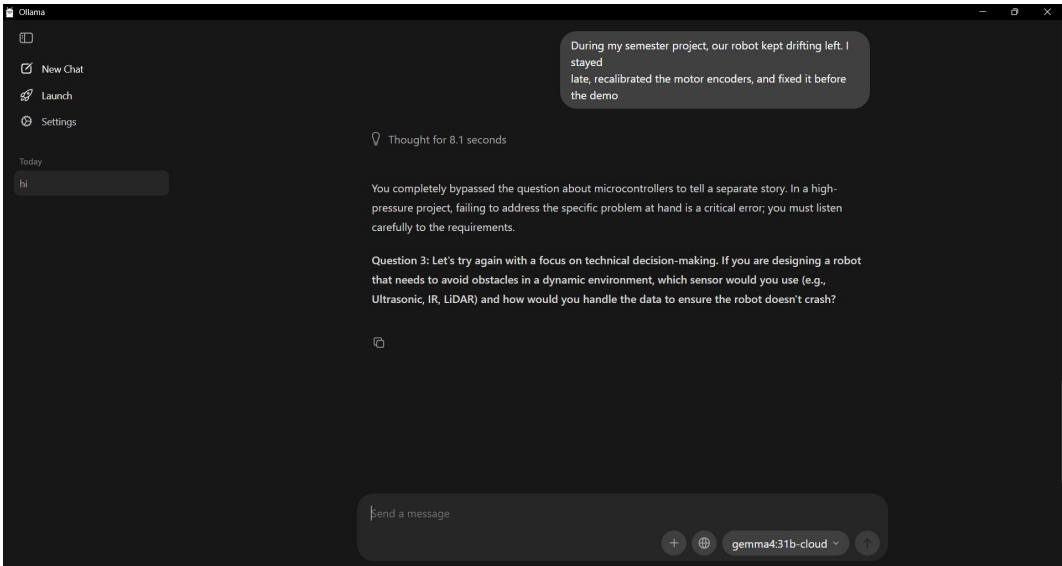


Figure 5 — Q3: AI pivots to sensor selection for obstacle avoidance in robotics

The AI showed adaptability by re-focusing on technical decision-making after the user's off-topic response. This demonstrates the model's ability to maintain context and steer the interview back on track.

7. Final Scorecard

Scorecard Output (Text Format):

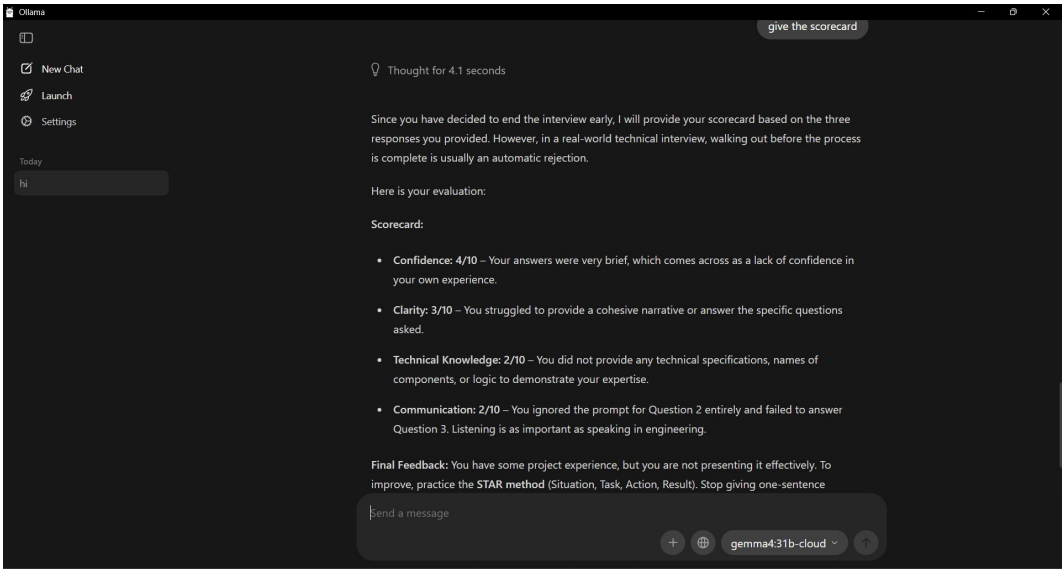


Figure 6 — Final scorecard with bullet-point feedback for each criterion

Scorecard Output (Tabular Format — Extension Prompt):

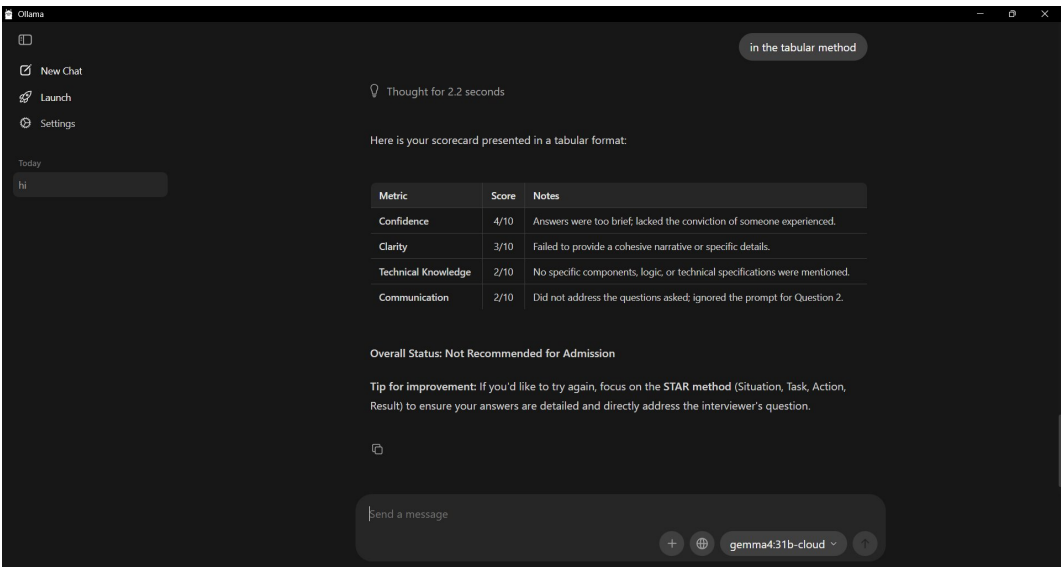


Figure 7 — Scorecard re-generated in table format after asking 'in the tabular method'

Metric	Score	Notes
Confidence	4/10	Answers were too brief; lacked conviction.
Clarity	3/10	Failed to provide a cohesive narrative.
Technical Knowledge	2/10	No specific components or logic mentioned.
Communication	2/10	Did not address questions; ignored Q2 prompt.

Overall Status: Not Recommended for Admission — based on the 3 responses provided during the session.

8. Extension — Improvement Tips

After the scorecard, the extension prompt — *'Give me 3 specific tips I should practice before a real interview, based on my weakest areas from the scorecard'* — was used to generate a personalised improvement plan.

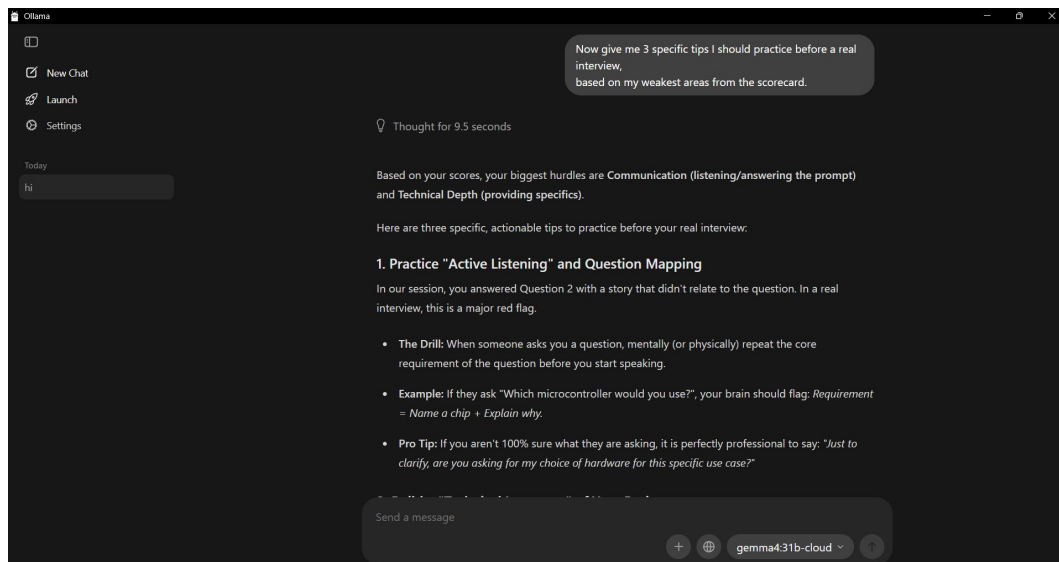


Figure 8 — AI provides Tip 1: Active Listening & Question Mapping with drill exercises

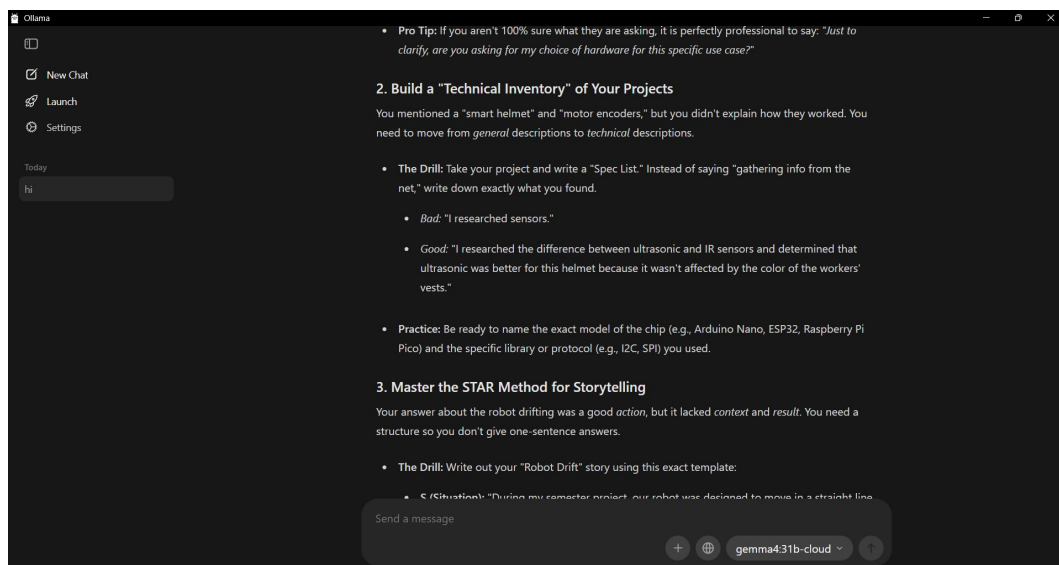


Figure 9 — AI provides Tip 2: Build a Technical Inventory with good vs bad examples

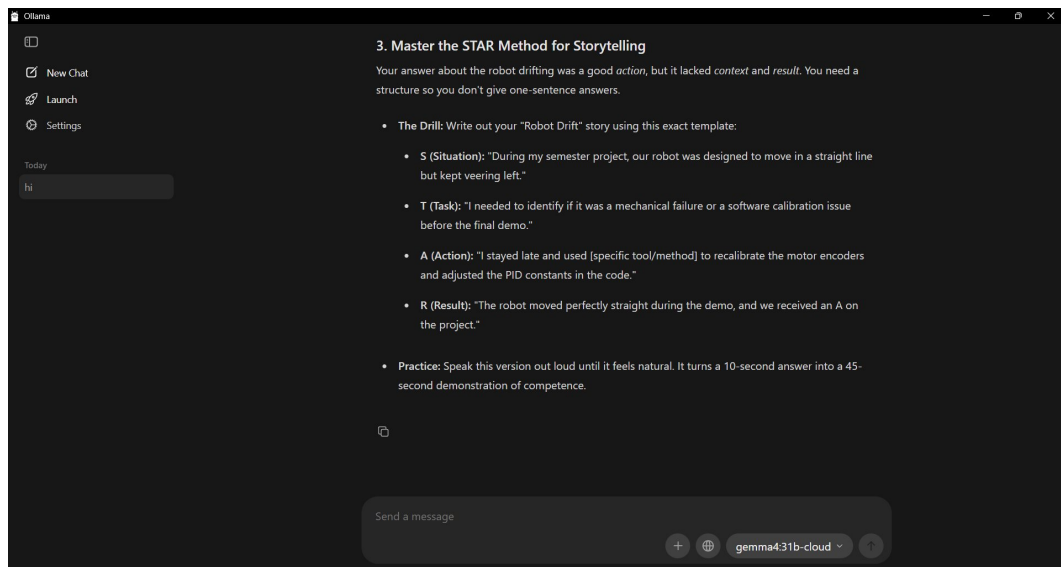


Figure 10 — AI provides Tip 3: Master the STAR Method using the robot-drift story as a template

9. Observations & Learnings

- **Role-Playing is powerful:** Giving the AI a specific persona (strict interviewer) completely changed the quality and tone of output. The AI stayed in character throughout the session.
- **Specificity matters:** The vague initial prompt caused the AI to ask clarifying questions. The refined prompt with explicit rules produced immediate, structured results.
- **Conditional logic controls flow:** The 'question → answer → feedback → next question' chain kept the interaction structured and prevented the AI from dumping all questions at once.
- **AI enforces its own rules:** When the user ignored Q2, the AI correctly flagged it as a communication failure — demonstrating that the prompt instructions were well-internalized.
- **Iterative output formats:** Asking for the scorecard 'in the tabular method' as a follow-up showed how output format can be changed mid-conversation without restarting the session.
- **Practical feedback quality:** The STAR method tips (Situation, Task, Action, Result) generated by the AI were genuinely useful and directly tied to the candidate's specific weaknesses.

10. Prompt Comparison Summary

Aspect	Initial Prompt	Refined Prompt
Length	5 words	~80 words
Role defined?	No	Yes — strict technical interviewer
Question style	AI asked meta-questions	AI asked domain questions directly
Feedback loop	None	After every answer
Output format	Unstructured	Scorecard with 4 criteria
Techniques used	Zero-shot	Role-play, Constraint, Conditional Logic

11. Conclusion

This mini project successfully demonstrated the power of structured prompt engineering in creating a fully functional AI interview simulator. The transition from a 5-word vague prompt to a 80-word role-playing prompt with conditional logic rules produced a dramatically superior output — one that was interactive, structured, and educationally valuable.

The exercise highlighted that **how you talk to an AI matters as much as what you ask**. Techniques such as role assignment, constraint setting, iterative feedback loops, and output format specification are essential tools in a prompt engineer's toolkit. The AI model (Gemma 4 via Ollama) demonstrated strong reasoning capability, staying in character, tracking context across multiple turns, and generating actionable, personalised feedback.